2. The linked genes exhibit some crossing over in complete linkage
3. The separation of two linked genes are possible in incomplete linkage
4. Crossing over is absent in complete linkage

a) 1 and 2 b) 2 and 3
c) 3 and 4 d) 1 and 4

5. Accurate mapping of genes can be done by three point test cross because increases

a) Possibility of single cross over
b) Possibility of double cross over
c) Possibility of multiple cross over
d) Possibility of recombination frequency

6. Due to incomplete linkage in maize, the ratio of parental and recombinants are

a) 50:50 b) 7:1:1:7 c) 96.4: 3.6 d) 1:7:7:1

7. Genes **G S L H** are located on same chromosome. The recombination percentage is between L and G is 15%, S and L is 50%, H and S are 20%. The correct order of genes is

a) GHSL b) SHGL c) SGHL d) HSLG

8. The point mutation sequence for transition, transition, transversion and transversion in DNA are

a) A to T, T to A, C to G and G to C
b) A to G, C to T, C to G and T to A
c) C to G, A to G, T to A and G to A
d) G to C, A to T, T to A and C to G

9. If haploid number in a cell is 18. The double monosomic and trisomic number will be

a) 35 and 37 b) 34 and 35
c) 37 and 35 d) 17 and 19

10. Changing the codon AGC to AGA represents

a) missense mutation b) nonsense mutation
c) frameshift mutation d) deletion mutation

11. **Assertion (A):** Gamma rays are generally use to induce mutation in wheat varieties.
Reason (R): Because they carry lower energy to non-ionize electrons from atom

a) A is correct. R is correct explanation of A
b) A is correct. R is not correct explanation of A
c) A is correct. R is wrong explanation of A
d) A and R is wrong

12. How many map units separate two alleles A and B if the recombination frequency is 0.09?

a) 900 cM b) 90 cM c) 9 cM d) 0.9 cM

13. Which one of the following pairs of codons is correctly matched with their function or the signal for the particular amino acid?

a) UUA, UCA - Leucine
b) GUU, GCU - Alanine
c) UAG, UGA - Stop
d) AUG, ACG – Start / Methionine

14. Removal of introns and joining of exons in a defined order during transcription is called

a) Splicing b) Looping
c) Inducing d) Slicing

15. If one strand of DNA has the nitrogenous base sequence as ATCTS, what would be the complementary RNA strand sequence?

a) ATCGU b) TTAGU
c) UAGAC d) AACTG

16. Removal of RNA polymerase III nucleoplasm will affect the synthesis of

a) rRNA b) tRNA
c) hnRNA d) mRNA

17. DNA dependent RNA polymerase catalyzes transcription on one strand of the DNA which is called the

a) Alpha strand b) Anti strand
c) Template strand d) Coding strand

18. Which of the following correctly represents the flow of genetic information?

a) DNA → RNA → Protein
b) RNA → DNA → Protein
c) RNA → Protein → DNA
d) Protein → RNA → DNA

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